

SHORT BIO.

My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University

Seoul, Republic of Korea

Integrated M.S. and Ph.D. in Data Science

Sep 2022 - Aug 2027 (expected)

Advisor: Prof. Sanghack Lee.

Overall GPA: 4.02 /4.3.

Department of Mathematics and Statistics, Sejong University

Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), Cum Laude

Mar 2016 - Aug 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

Min Woo Park and Sanghack Lee†. Counterfactual Structural Causal Bandits. *International Conference on Learning Representations* (ICLR), 2026.

Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. Structural Causal Bandits under Markov Equivalence. *Neural Information Processing Systems* (NeurIPS), 2025, Columbia CausalAI Laboratory, Technical Report (R-122)

Min Woo Park and Sanghack Lee†. On Transportability for Structural Causal Bandits. *arXiv preprint*, 2025.

Yonghan Jung†, Min Woo Park, and Sanghack Lee†. Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference. *Neural Information Processing Systems* (NeurIPS), 2024.

Min Woo Park*, and Ji Eun Lee*†. Computation of the iterated Aluthge, Duggal, and Mean transforms. *Filomat*, 2023.

Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University Spring 2024, Spring 2025, Spring 2026

BK21 scholarship

Fall 2024, Fall 2025

First place merit scholarship | Sejong University

Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University

Nov 2021

▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University

Nov 2020

▷ The significance of the Euler's number and its applications.

PROJECTS

- Towards More Scalable Score-based Causal Discovery | LG AI Research Sep 2025 - Aug 2026
- ▷ Working on developing scalable score-based causal discovery algorithms.
 - To be submitted to *Uncertainty in Artificial Intelligence (UAI) 2026* as first author.
- Hyundai NGV Data Boot-Campus | Hyundai Motors July 2025 - Sep 2025
- ▷ Development of high-voltage battery performance prediction technology.
- Hyundai NGV Data Boot-Campus | Hyundai Motors July 2024 - Sep 2024
- ▷ Development of an intelligent system for predicting vulnerable regions of automotive body panel stiffness through differential geometry-based automatic curvature analysis.

TEACHING ASSISTENTS

- Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026
- ▷ Bayesian Networks and Pearl's Graphical Causal Inference Framework.
- Machine Learning and Deep Learning for Data Science II Fall 2023
- ▷ Kernel Methods, Gaussian Processes, and Related Topics for Machine Learning
- Basic Mathematics for Artificial Intelligence Spring 2022
- ▷ Linear Algebra, Basic Statistics, and Probability Theory for Artificial Intelligence.

PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

- ICLR 2026 poster session | Rio de Janeiro, Brazil Apr 2026 (expected)

Structural Causal Bandits under Markov Equivalence

- ExploreCSR poster session | Seoul, Republic of Korea Dec 2025
- NeurIPS 2025 poster session | San Diego, USA Dec 2025
- JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea Nov 2025
- SNU GSDS student research seminar | Seoul, Republic of Korea Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

- NeurIPS 2024 poster session | Vancouver, Canada Dec 2024
- SNU GSDS student research seminar | Seoul, Republic of Korea Dec 2024

ACADEMIC SERVICES

Conference Reviewer: 2026 ICLR, ICML, 2025 ICML

REFERENCE

- Sanghack Lee** | Seoul National University sanghack@snu.ac.kr
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Associate professor
- Ji Eun Lee** | Sejong University jieunlee7@sejong.ac.kr
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- Jin Hee Yoon** | Sejong University jin9135@sejong.ac.kr
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